2) Pole shoes spread out the flux in the air gap uniformly.
3) They support the exciting coils (or field coils).
4) It is made up of magnetic material like cast-iron or cast-steel.

ii) Field Winding

The field winding is placed on the pole core. It is made up of aluminium or copper. Their functions are to carry the current to produce the required magnetic flux. The pole structure of a DC machines is illustrated in fig 7.4.

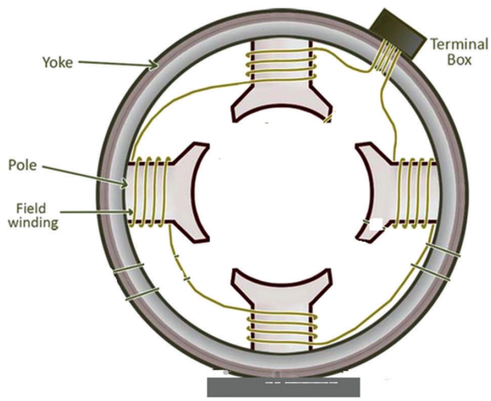


Fig 7.4 *Yoke and Poles*

7.3.3 Armature System

It is further divided into two parts namely:

1) Armature core
2)) Armature winding

i) Armature Core

It is cylindrical in shape with slots on its outer periphery. It is mounted on the shaft. They are used to house the armature conductors in the slot. It is made of cast-steel laminations to reduce the eddy current loss.

ii) Armature Slots and Windings

The armature windings are placed into the slots on the armature surface. When the armature rotates, an emf is induced in the armature conductors in case of generators. The ends of the coils are soldered with commutator segments. These segments are made up of copper materials.

7.3.4 Commutator

The emf induced in the armature conductor is alternating in nature. This alternating emf is made into unidirectional by means of a device called as commutator. It is made up of copper segments.

7.3.5 Brushes and Bearings

The function of brushes is to collect current from commutator. They are usually made up of carbon or graphite and in the shape of rectangular block. These brushes are housed in brush-holders usually of the box-type variety. Ball bearings are frequently employed because of their reliability. But for heavy duty machines, roller bearing is preferable. The complete structure is shown in fig 7.5.

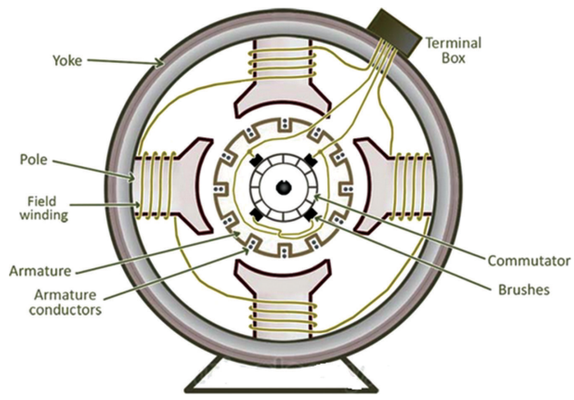


Fig 7.5 *Poles and Armature of DC Machines.*

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